

Chapter 6

The Obidi Convention

Hierarchical Tensor Structures and Sectoral Indexing

6.1 Introduction

The Hybrid Manifold Architecture System (HMAS) introduces a fundamentally new mathematical challenge.

Classical tensor calculus was developed for geometric structures possessing a single metric and a single geometrical sector. In General Relativity, every tensor lives on one Lorentzian manifold and all index operations are performed relative to a single metric tensor.

The Theory of Entropicity departs from this assumption.

The Hybrid Manifold contains multiple interacting geometrical sectors:

$$FR, \quad FS, \quad L.$$

Each sector possesses its own metric structure, curvature tensors, and geometric dynamics.

Consequently, a new indexing framework becomes necessary.

The Obidi Convention provides this framework.

Its purpose is to extend tensor notation so that multiple geometrical sectors may be treated simultaneously while preserving mathematical consistency.

The Obidi Convention therefore constitutes the formal language of the Theory of Entropicity.

6.2 Motivation

The necessity of the Obidi Convention may be understood through a simple example.

Consider the metric decomposition

$$g_{\mu\nu} = g_{\mu\nu}^{(FR)} + g_{\mu\nu}^{(FS)} + g_{\mu\nu}^{(L)}.$$

Each term belongs to a different geometrical sector.

Using ordinary tensor notation, there is no intrinsic mechanism for distinguishing these sectors once tensor manipulations become complicated.

Questions immediately arise:

- Which metric is being used to raise an index?
- Which curvature tensor is being contracted?
- Which sector participates in a given variation?
- Which geometric contribution enters the action?

The Obidi Convention was developed to answer these questions systematically.

6.3 Fundamental Principle

The fundamental principle of the Obidi Convention is:

Every tensor possesses both a geometric coordinate identity and a sectoral identity.

The geometric identity specifies position within a manifold.

The sectoral identity specifies membership within a particular geometric sector.

Consequently every tensor index becomes hierarchical.

6.4 Primary Indices

Definition

A primary index is an ordinary tensor index describing coordinates on the underlying manifold.

These indices are denoted by Greek letters:

$$\mu, \nu, \rho, \sigma, \lambda, \dots$$

with

$$\mu = 0, 1, 2, 3.$$

The primary index retains exactly the same interpretation as in conventional differential geometry.

For example:

$$V^\mu$$

denotes a contravariant vector,

while

$$T_{\mu\nu}$$

denotes a rank-two covariant tensor.

The introduction of the Obidi Convention does not alter the meaning of primary indices.

It supplements them.

6.5 Secondary Indices

Definition

A secondary index identifies the sector to which a tensor belongs.

The secondary index is called the Obidi Index.

It is denoted by

$$J.$$

The admissible values are

$$J \in \{FR, FS, L\}.$$

Explicitly,

$$FR = \text{Fisher--Rao Sector},$$

$$FS = \text{Fubini--Study Sector},$$

$$L = \text{Lorentzian Sector}.$$

The secondary index therefore specifies the geometric origin of a tensor.

6.6 Definition of the Obidi Index

Formal Definition

The Obidi Index is the secondary hierarchical index attached to every tensor in HMAS.

A tensor carrying an Obidi Index belongs to a specific sector.

For example,

$$T_{(\mu)FR}$$

denotes the Fisher–Rao component of a vector.

Similarly,

$$T_{(\mu)FS}$$

denotes the Fubini–Study component,

while

$$T_{(\mu)L}$$

denotes the Lorentzian component.

The Obidi Index therefore functions as a sectoral coordinate.

6.7 Axiom I

Principle of Sectoral Identity

Every tensor belongs to exactly one sector before any addition operation is performed.

Symbolically,

$$T \in FR \quad \text{or} \quad FS \quad \text{or} \quad L.$$

No tensor is simultaneously a member of multiple sectors.

Sectoral combinations arise only through the Obidi Addition Rule.

6.8 The Obidi Bracket Rule

Motivation

A tensor now carries two types of indices.

To avoid ambiguity, a standardized notation is required.

The Obidi Bracket Rule provides this notation.

Formal Statement

Whenever a primary index carries a secondary index, the primary index shall be enclosed within brackets.

Examples:

$$T_{(\mu)J},$$

$$V^{(\mu)J},$$

$$g_{(\mu\nu)J},$$

$$G_{(\mu\nu)J}.$$

The brackets visually indicate that the sectoral label belongs to the indexed quantity.

Interpretation

The notation

$$T_{(\mu)FR}$$

means:

"the component of tensor T associated with coordinate index μ in the Fisher–Rao sector."

Likewise,

$$g_{(\mu\nu)L}$$

means:

"the metric tensor in the Lorentzian sector."

This notation eliminates ambiguity throughout the theory.

6.9 Axiom II

Principle of Hierarchical Indexing

Every tensor in HMAS possesses a two-level indexing structure:

(Primary Index, Secondary Index).

Symbolically,

$$(\mu, J).$$

The primary index determines geometric coordinates.

The secondary index determines sectoral membership.

6.10 Sectoral Tensor Decomposition

The Obidi Convention permits every physical tensor to be decomposed into sectoral components.

For a vector,

$$T_\mu = T_{(\mu)FR} + T_{(\mu)FS} + T_{(\mu)L}.$$

For a rank-two tensor,

$$T_{\mu\nu} = T_{(\mu\nu)FR} + T_{(\mu\nu)FS} + T_{(\mu\nu)L}.$$

For the metric,

$$g_{\mu\nu} = g_{(\mu\nu)FR} + g_{(\mu\nu)FS} + g_{(\mu\nu)L}.$$

This decomposition provides the foundation for all subsequent calculations.

6.11 Axiom III

Principle of Sectoral Decomposition

Every observable tensor may be expressed as the sum of its sectoral components.

Formally,

$$T = \sum_J T^{(J)}.$$

This principle underlies the Obidi Addition Rule.

6.12 Obidi Orthogonality Principle

One of the most important features of the Obidi Convention is sectoral orthogonality.

Different sectors are regarded as geometrically distinct.

Consequently they do not contract directly.

Formal Statement

Let

$$T^{(\mu)J}$$

and

$$S_{(\mu)K}$$

belong to sectors J and K .

Then

$$T^{(\mu)J} S_{(\mu)K} = 0, \quad J \neq K.$$

This is called Obidi Orthogonality.

Interpretation

The Fisher–Rao sector does not directly contract with the Fubini–Study sector.

The Fubini–Study sector does not directly contract with the Lorentzian sector.

The Lorentzian sector does not directly contract with the Fisher–Rao sector.

Only matching sectors participate in contractions.

6.13 Axiom IV

Principle of Sectoral Orthogonality

Distinct sectors are orthogonal under contraction.

Formally,

$$FR \perp FS,$$

$$FR \perp L,$$

$$FS \perp L.$$

This principle preserves mathematical consistency across HMAS.

6.14 Einstein–Obidi Summation Convention

The introduction of hierarchical indices requires a corresponding extension of Einstein summation.

Standard Einstein Summation

In conventional tensor calculus,

$$A^\mu B_\mu = \sum_{\mu=0}^3 A^\mu B_\mu.$$

The repeated index implies summation.

Obidi Generalization

In HMAS, summation occurs only when both:

1. Primary indices match.
2. Secondary indices match.

Thus

$$A^{(\mu)J} B_{(\mu)J} = \sum_{\mu=0}^3 A^{(\mu)J} B_{(\mu)J}.$$

This contraction is valid.

However,

$$A^{(\mu)FR} B_{(\mu)FS} = 0.$$

The mismatch in sectoral labels prevents contraction.

6.15 Axiom V

Einstein–Obidi Summation Principle

A repeated index pair

$$(\mu, J)$$

is summed only when both indices coincide.

Symbolically,

$$A^{(\mu)J} B_{(\mu)K} = \delta_K^J A^{(\mu)J} B_{(\mu)J}.$$

The Kronecker delta

$$\delta_K^J$$

enforces sectoral matching.

6.16 Sectoral Kronecker Delta

To formalize orthogonality, introduce the sectoral Kronecker delta

$$\delta_K^J = \begin{cases} 1, & J = K, \\ 0, & J \neq K. \end{cases}$$

This object functions analogously to the ordinary Kronecker delta but acts on sector labels.

The orthogonality relation becomes

$$T^{(\mu)J} S_{(\mu)K} = \delta_K^J T^{(\mu)J} S_{(\mu)J}.$$

6.17 Sectoral Metric Tensors

Each sector possesses its own metric.

These are

$$\begin{aligned} g_{\mu\nu}^{(FR)}, \\ g_{\mu\nu}^{(FS)}, \\ g_{\mu\nu}^{(L)}. \end{aligned}$$

Indices may only be raised and lowered within their own sectors.

For example,

$$V_{(\mu)FR} = g_{\mu\nu}^{(FR)} V^{(\nu)FR}.$$

Cross-sector index operations are forbidden.

6.18 Sectoral Curvature Objects

Each sector possesses its own curvature hierarchy:

$$\begin{aligned} R_{\sigma\mu\nu}^{(J)}, \\ R_{\mu\nu}^{(J)}, \\ R^{(J)}, \\ G_{\mu\nu}^{(J)}. \end{aligned}$$

These quantities are computed independently within each sector.

No mixed-sector curvature tensors are assumed at the level of the basic Obidi Convention.

6.19 Conceptual Significance

The Obidi Convention accomplishes more than notational organization.

It provides a rigorous mathematical language capable of representing:

- classical information geometry,
- quantum information geometry,
- spacetime geometry,

within a single tensor framework.

This achievement is essential for the construction of the Local Obidi Action.

Without hierarchical indexing, the theory would lack a consistent mathematical foundation.

6.20 Transition to Obidi Calculus

Having established the indexing framework, the next step is to define the algebra governing these objects.

The following chapter introduces Obidi Calculus, which specifies:

- Addition Rules,
- Multiplication Rules,
- Contraction Rules,
- Sectoral Products,
- Determinant Factorization,
- Partition Function Factorization,

and the mathematical operations required for constructing the Local Obidi Action and the Obidi Field Equations.

The Obidi Convention provides the language.

Obidi Calculus provides the grammar.